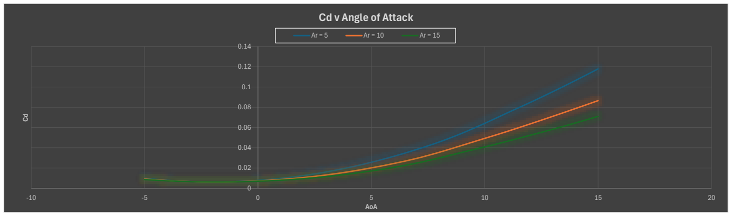
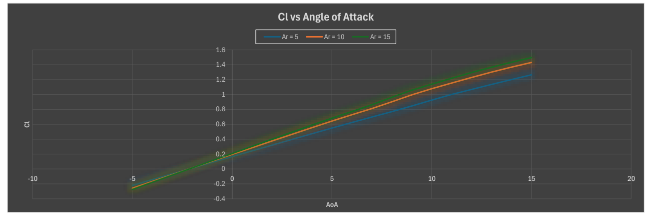
## Rectangular Wing

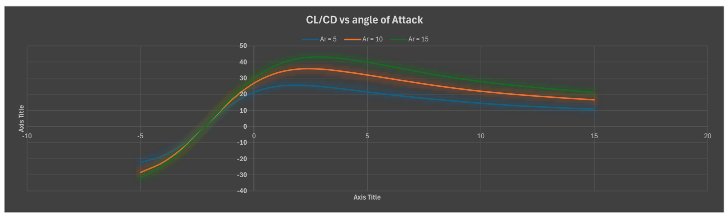
| AR = 5 | | | | |
|---|---|---|---|---|
| α | Cd | Cl | Cl/Cd | e |
| -5 | 0.009860878 | -0.220404 | -22.35136 | 0.9737251 |
| -4 | 0.007849018 | -0.1424054 | -18.14308 | 0.9507569 |
| -3 | 0.006619826 | -0.06548542 | -9.892317 | 0.9362252 |
| -2 | 0.006198295 | 0.01160394 | 1.872119 | 1.163814 |
| -1 | 0.006598236 | 0.08872265 | 13.44642 | 0.9866772 |
| 0 | 0.007795724 | 0.1658969 | 21.2805 | 0.976332 |
| 1 | 0.009773226 | 0.2430072 | 24.86458 | 0.9725264 |
| 2 | 0.0124996 | 0.319854 | 25.58913 | 0.9704395 |
| 3 | 0.01605314 | 0.3967314 | 24.71363 | 0.9692602 |
| 4 | 0.02044053 | 0.4732686 | 23.15344 | 0.9682813 |
| 5 | 0.02563692 | 0.549183 | 21.42156 | 0.9673622 |
| 6 | 0.03160844 | 0.6241354 | 19.74585 | 0.9664006 |
| 7 | 0.03832093 | 0.697725 | 18.20741 | 0.9651182 |
| 8 | 0.04573491 | 0.7695748 | 16.82686 | 0.9639389 |
| 9 | 0.05434789 | 0.845607 | 15.55915 | 0.9648834 |
| 10 | 0.06409585 | 0.9241337 | 14.41799 | 0.9658225 |
| 11 | 0.07449499 | 1.000463 | 13.42994 | 0.965449 |
| 12 | 0.08469144 | 1.068597 | 12.61753 | 0.9629091 |
| 13 | 0.09536708 | 1.135704 | 11.90877 | 0.9612854 |
| 14 | 0.1064506 | 1.20129 | 11.28495 | 0.959905 |
| 15 | 0.117901 | 1.264446 | 10.72464 | 0.9576019 |

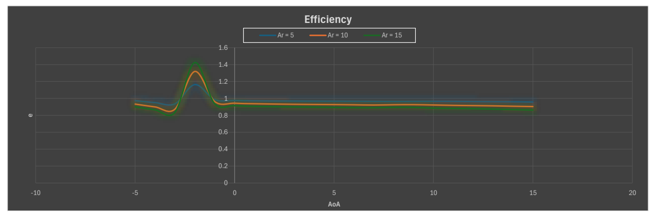
| AR = 10 | | | | |
|---|---|---|---|---|
| α | Cd | Cl | Cl/Cd | e |
| -5 | 0.009066535 | -0.2584113 | -28.50166 | 0.9335376 |
| -4 | 0.007534532 | -0.1670497 | -22.17122 | 0.9000166 |
| -3 | 0.006563081 | -0.07678297 | -11.69923 | 0.8739308 |
| -2 | 0.00619174 | 0.01368671 | 2.210478 | 1.319043 |
| -1 | 0.006429718 | 0.1042873 | 16.21957 | 0.9637544 |
| 0 | 0.007251609 | 0.1949005 | 26.87686 | 0.9434929 |
| 1 | 0.008606132 | 0.2853051 | 33.15138 | 0.9375744 |
| 2 | 0.01053408 | 0.3754978 | 35.64601 | 0.9340098 |
| 3 | 0.01310097 | 0.4654323 | 35.52656 | 0.9311352 |
| 4 | 0.01630608 | 0.55457 | 34.01002 | 0.929151 |
| 5 | 0.02011735 | 0.6422712 | 31.92624 | 0.9274744 |
| 6 | 0.02451575 | 0.7278534 | 29.68921 | 0.925121 |
| 7 | 0.02950289 | 0.8107488 | 27.48032 | 0.9221874 |
| 8 | 0.03554312 | 0.9024999 | 25.39169 | 0.9256902 |
| 9 | 0.04235049 | 0.9949793 | 23.49393 | 0.9266776 |
| 10 | 0.04927072 | 1.077965 | 21.87842 | 0.9226011 |
| 11 | 0.05618525 | 1.153714 | 20.53412 | 0.9181639 |
| 12 | 0.06353084 | 1.22948 | 19.35248 | 0.9152425 |
| 13 | 0.07104234 | 1.301408 | 18.31877 | 0.9123742 |
| 14 | 0.07872312 | 1.36823 | 17.38029 | 0.9078655 |
| 15 | 0.08653932 | 1.430781 | 16.53331 | 0.9039596 |

| AR = 15 | | | | |
|---|---|---|---|---|
| α | Cd | Cl | Cl/Cd | e |
| -5 | 0.008615434 | -0.2748799 | -31.90552 | 0.9010381 |
| -4 | 0.00734925 | -0.1778974 | -24.2062 | 0.8647391 |
| -3 | 0.006527475 | -0.08183598 | -12.53716 | 0.8353512 |
| -2 | 0.006188881 | 0.01451656 | 2.345588 | 1.424434 |
| -1 | 0.006340792 | 0.110914 | 17.49214 | 0.9366055 |
| 0 | 0.006954841 | 0.2074363 | 29.82618 | 0.9148122 |
| 1 | 0.007970269 | 0.3035353 | 38.08345 | 0.9065487 |
| 2 | 0.009459836 | 0.3995335 | 42.23471 | 0.9025232 |
| 3 | 0.01148305 | 0.495101 | 43.11582 | 0.8996721 |
| 4 | 0.01404 | 0.5894377 | 41.98274 | 0.8970156 |
| 5 | 0.01710493 | 0.6818273 | 39.86144 | 0.8941697 |
| 6 | 0.02067782 | 0.7714666 | 37.30889 | 0.8911112 |
| 7 | 0.02486056 | 0.861785 | 34.66474 | 0.890181 |
| 8 | 0.02993845 | 0.9618856 | 32.12877 | 0.894953 |
| 9 | 0.03545326 | 1.057764 | 29.83546 | 0.8941894 |
| 10 | 0.0408223 | 1.13793 | 27.87521 | 0.8854889 |
| 11 | 0.04642099 | 1.217723 | 26.23216 | 0.8819736 |
| 12 | 0.05230102 | 1.295884 | 24.77742 | 0.8790451 |
| 13 | 0.05825809 | 1.366843 | 23.46185 | 0.8739669 |
| 14 | 0.06446227 | 1.433665 | 22.24037 | 0.8685456 |
| 15 | 0.07096774 | 1.495779 | 21.07689 | 0.862767 |




**CL Vs. AoA**

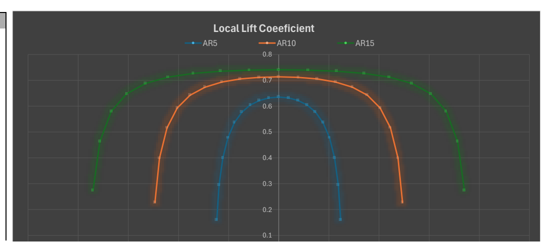
| | | |
|---|---|---|
| **Wing 1 experimental** | **Wing 2 experiment** | **Wing 3 experimental** |
| 4.295268674 | 4.929001383 | 5.185070489 |
| **Wing 1 theory** | **Wing 2 theory** | **Wing 3 theory** |
| 4.487989505 | 5.235987756 | 5.543987036 |
| **Wing 1 % error** | **Wing 2 % error** | **Wing 3 % error** |
| 4.29415% | 5.86301% | 6.47398% |

**5 dgerees AoA**

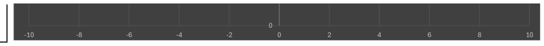
| | AR=5 | AR=10 | AR=15 |
|---|---|---|---|
| **Total Lift Coeff (CL)** | 0.549 | 0.642 | 0.682 |
| **Oswald Efficiency €** | 0.967 | 0.927 | 0.894 |
| **Induced Angle ai (rad)** | 0.034950426 | 0.020435495 | 0.014472489 |
| **Induced Angle ai (deg)** | 2.002511874 | 1.170867598 | 0.829212567 |
| **Effective Angle (deg)** | 2.997488126 | 3.829132402 | 4.170787433 |
| **Downwash Vel (m/s)** | 1.747165521 | 1.021703619 | 0.723599214 |
| **Induced Drag Coeff** | 0.019842589 | 0.014152737 | 0.011040534 |

TOTAL LIFT COEEFICIENT AT 5 degrees

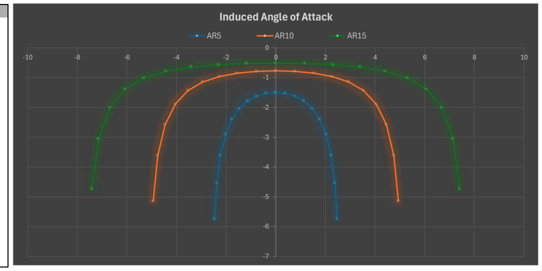
| AR=5 | | | AR=10 | | | AR=15 | |
|---|---|---|---|---|---|---|---|
| 2y/b | Cl | | 2y/b | Cl | | 2y/b | Cl |
| -2.469221 | 0.1598889 | | -4.938442 | 0.2269799 | | -7.407663 | 0.2736436 |
| -2.377641 | 0.2947316 | | -4.755283 | 0.3990752 | | -7.132924 | 0.4635286 |
| -2.227516 | 0.3999648 | | -4.455033 | 0.5164359 | | -6.682549 | 0.5796919 |
| -2.022542 | 0.4787222 | | -4.045085 | 0.593133 | | -6.067627 | 0.6481408 |
| -1.767767 | 0.535906 | | -3.535534 | 0.642198 | | -5.303301 | 0.6884773 |
| -1.469463 | 0.5764383 | | -2.938926 | 0.6735307 | | -4.408389 | 0.7123225 |
| -1.134976 | 0.6041601 | | -2.269952 | 0.6932795 | | -3.404929 | 0.7266128 |
| -0.7725425 | 0.6220549 | | -1.545085 | 0.7051613 | | -2.317627 | 0.7350531 |
| -0.3910862 | 0.6320736 | | -0.7821723 | 0.7115294 | | -1.173258 | 0.7395081 |
| -4.04E-15 | 0.6352982 | | -8.08E-15 | 0.713541 | | -1.21E-14 | 0.7409086 |
| 0.3910862 | 0.6320736 | | 0.7821723 | 0.7115294 | | 1.173258 | 0.7395081 |
| 0.7725425 | 0.6220549 | | 1.545085 | 0.7051613 | | 2.317627 | 0.7350531 |
| 1.134976 | 0.6041601 | | 2.269952 | 0.6932795 | | 3.404929 | 0.7266128 |
| 1.469463 | 0.5764383 | | 2.938926 | 0.6735307 | | 4.408389 | 0.7123225 |
| 1.767767 | 0.535906 | | 3.535534 | 0.642198 | | 5.303301 | 0.6884773 |
| 2.022542 | 0.4787222 | | 4.045085 | 0.593133 | | 6.067627 | 0.6481408 |



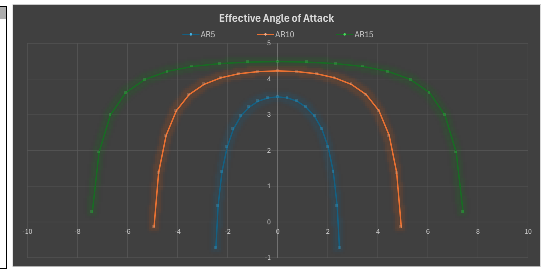
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2.227516 | 0.3999648 | | 4.455033 | 0.5164359 | | 6.682549 | 0.5796919 |
| 2.377641 | 0.2947316 | | 4.755283 | 0.3990752 | | 7.132924 | 0.4635286 |
| 2.469221 | 0.1598889 | | 4.938442 | 0.2269799 | | 7.407663 | 0.2736436 |



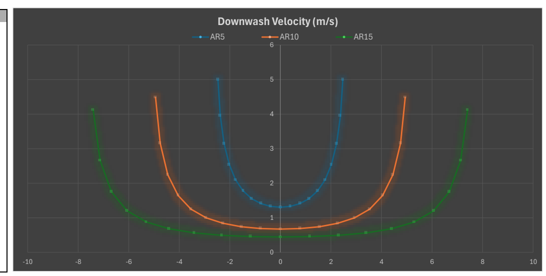
| AR=5 | | | AR=10 | | | AR=15 | |
|---|---|---|---|---|---|---|---|
| 2y/b | Induced AoA | | 2y/b | Induced AoA | | 2y/b | Induced AoA |
| -2.469221 | -5.733903 | | -4.938442 | -5.143405 | | -7.407663 | -4.730678 |
| -2.377641 | -4.543672 | | -4.755283 | -3.618919 | | -7.132924 | -3.048184 |
| -2.227516 | -3.611041 | | -4.455033 | -2.574764 | | -6.682549 | -2.008066 |
| -2.022542 | -2.912633 | | -4.045085 | -1.884899 | | -6.067627 | -1.38003 |
| -1.767767 | -2.400335 | | -3.535534 | -1.434574 | | -5.303301 | -1.009816 |
| -1.469463 | -2.037214 | | -2.938926 | -1.146997 | | -4.408389 | -0.7832708 |
| -1.134976 | -1.783691 | | -2.269952 | -0.9644798 | | -3.404929 | -0.6472866 |
| -0.7725425 | -1.61945 | | -1.545085 | -0.8514146 | | -2.317627 | -0.5669701 |
| -0.3910862 | -1.527497 | | -0.7821723 | -0.790817 | | -1.173258 | -0.5245776 |
| -4.04E-15 | -1.497901 | | -8.08E-15 | -0.7716758 | | -1.21E-14 | -0.511251 |
| 0.3910862 | -1.527497 | | 0.7821723 | -0.790817 | | 1.173258 | -0.5245776 |
| 0.7725425 | -1.61945 | | 1.545085 | -0.8514146 | | 2.317627 | -0.5669701 |
| 1.134976 | -1.783691 | | 2.269952 | -0.9644798 | | 3.404929 | -0.6472866 |
| 1.469463 | -2.037214 | | 2.938926 | -1.146997 | | 4.408389 | -0.7832708 |
| 1.767767 | -2.400335 | | 3.535534 | -1.434574 | | 5.303301 | -1.009816 |
| 2.022542 | -2.912633 | | 4.045085 | -1.884899 | | 6.067627 | -1.38003 |
| 2.227516 | -3.611041 | | 4.455033 | -2.574764 | | 6.682549 | -2.008066 |
| 2.377641 | -4.543672 | | 4.755283 | -3.618919 | | 7.132924 | -3.048184 |
| 2.469221 | -5.733903 | | 4.938442 | -5.143405 | | 7.407663 | -4.730678 |




| AR=5 | | | AR=10 | | | AR=15 | |
|---|---|---|---|---|---|---|---|
| 2y/b | effective angle | | 2y/b | effective angle | | 2y/b | effective angle |
| -2.469221 | -0.7339032 | | -4.938442 | -0.1434047 | | -7.407663 | 0.2693216 |
| -2.377641 | 0.4563282 | | -4.755283 | 1.381081 | | -7.132924 | 1.951816 |
| -2.227516 | 1.388959 | | -4.455033 | 2.425236 | | -6.682549 | 2.991934 |
| -2.022542 | 2.087367 | | -4.045085 | 3.115101 | | -6.067627 | 3.61997 |
| -1.767767 | 2.599665 | | -3.535534 | 3.565426 | | -5.303301 | 3.990184 |
| -1.469463 | 2.962786 | | -2.938926 | 3.853003 | | -4.408389 | 4.216729 |
| -1.134976 | 3.216309 | | -2.269952 | 4.03552 | | -3.404929 | 4.352713 |
| -0.7725425 | 3.38055 | | -1.545085 | 4.148585 | | -2.317627 | 4.43303 |
| -0.3910862 | 3.472503 | | -0.7821723 | 4.209183 | | -1.173258 | 4.475422 |
| -4.04E-15 | 3.502099 | | -8.08E-15 | 4.228324 | | -1.21E-14 | 4.488749 |
| 0.3910862 | 3.472503 | | 0.7821723 | 4.209183 | | 1.173258 | 4.475422 |
| 0.7725425 | 3.38055 | | 1.545085 | 4.148585 | | 2.317627 | 4.43303 |
| 1.134976 | 3.216309 | | 2.269952 | 4.03552 | | 3.404929 | 4.352713 |
| 1.469463 | 2.962786 | | 2.938926 | 3.853003 | | 4.408389 | 4.216729 |
| 1.767767 | 2.599665 | | 3.535534 | 3.565426 | | 5.303301 | 3.990184 |
| 2.022542 | 2.087367 | | 4.045085 | 3.115101 | | 6.067627 | 3.61997 |
| 2.227516 | 1.388959 | | 4.455033 | 2.425236 | | 6.682549 | 2.991934 |
| 2.377641 | 0.4563282 | | 4.755283 | 1.381081 | | 7.132924 | 1.951816 |
| 2.469221 | -0.7339032 | | 4.938442 | -0.1434047 | | 7.407663 | 0.2693216 |




| AR=5 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2y/b | | | AR=10 | | | AR=15 | |
| -2.469221 | Downwash Velocity (m/s) | | 2y/b | Downwash Velocity (m/s) | | 2y/b | Downwash Velocity (m/s) |
| -2.377641 | 4.995426279 | | -4.938442 | 4.482441616 | | -7.407663 | 4.123606426 |
| -2.227516 | 3.960947169 | | -4.755283 | 3.156003168 | | -7.132924 | 2.658787728 |
| -2.022542 | 3.149141996 | | -4.455033 | 2.246149293 | | -6.682549 | 1.752009443 |
| -1.767767 | 2.540657196 | | -4.045085 | 1.644587999 | | -6.067627 | 1.204186923 |
| -1.469463 | 2.094074773 | | -3.535534 | 1.251771184 | | -5.303301 | 0.88118508 |
| -1.134976 | 1.777430023 | | -2.938926 | 1.000876854 | | -4.408389 | 0.683511429 |
| -0.7725425 | 1.556312626 | | -2.269952 | 0.841627655 | | -3.404929 | 0.564852103 |
| -0.3910862 | 1.413048565 | | -1.545085 | 0.742972059 | | -2.317627 | 0.494766676 |
| -4.04E-15 | 1.3328347 | | -0.7821723 | 0.69009611 | | -1.173258 | 0.45777392 |
| 0.3910862 | 1.307016319 | | -8.08E-15 | 0.673393815 | | -1.21E-14 | 0.446144742 |
| 0.7725425 | 1.3328347 | | 0.7821723 | 0.69009611 | | 1.173258 | 0.45777392 |
| 1.134976 | 1.413048565 | | 1.545085 | 0.742972059 | | 2.317627 | 0.494766676 |
| 1.469463 | 1.556312626 | | 2.269952 | 0.841627655 | | 3.404929 | 0.564852103 |
| 1.767767 | 1.777430023 | | 2.938926 | 1.000876854 | | 4.408389 | 0.683511429 |
| 2.022542 | 2.094074773 | | 3.535534 | 1.251771184 | | 5.303301 | 0.88118508 |
| 2.227516 | 2.540657196 | | 4.045085 | 1.644587999 | | 6.067627 | 1.204186923 |
| 2.377641 | 3.149141996 | | 4.455033 | 2.246149293 | | 6.682549 | 1.752009443 |
| 2.469221 | 3.960947169 | | 4.755283 | 3.156003168 | | 7.132924 | 2.658787728 |
| | 4.995426279 | | 4.938442 | 4.482441616 | | 7.407663 | 4.123606426 |




| AR=5 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2y/b | | | | | | | |
| -2.469221 | | | | | | | |
| -2.377641 | | | AR=10 | | | AR=15 | |
| -2.227516 | Induced Drag Coeff. | | 2y/b | Induced Drag Coeff. | | 2y/b | Induced Drag Coeff. |
| -2.022542 | 0.01600096 | | -4.938442 | 0.02037584 | | -7.407663 | 0.02259364 |
| -1.767767 | 0.02337282 | | -4.755283 | 0.02520641 | | -7.132924 | 0.02466012 |
| -1.469463 | 0.02520761 | | -4.455033 | 0.02320765 | | -6.682549 | 0.02031667 |
| -1.134976 | 0.02433586 | | -4.045085 | 0.01951271 | | -6.067627 | 0.01561116 |
| -0.7725425 | 0.02245111 | | -3.535534 | 0.01607938 | | -5.303301 | 0.01213414 |
| -0.3910862 | 0.0204959 | | -2.938926 | 0.01348333 | | -4.408389 | 0.009737914 |
| -4.04E-15 | 0.01880828 | | -2.269952 | 0.01167022 | | -3.404929 | 0.008208751 |
| 0.3910862 | 0.01758221 | | -1.545085 | 0.01047869 | | -2.317627 | 0.007273714 |
| 0.7725425 | 0.01685099 | | -0.7821723 | 0.009820786 | | -1.173258 | 0.006770645 |
| 1.134976 | 0.0166088 | | -8.08E-15 | 0.009610172 | | -1.21E-14 | 0.006611136 |
| 1.469463 | 0.01685099 | | 0.7821723 | 0.009820786 | | 1.173258 | 0.006770645 |
| 1.767767 | 0.01758221 | | 1.545085 | 0.01047869 | | 2.317627 | 0.007273714 |
| 2.022542 | 0.01880828 | | 2.269952 | 0.01167022 | | 3.404929 | 0.008208751 |
| 2.227516 | 0.0204959 | | 2.938926 | 0.01348333 | | 4.408389 | 0.009737914 |
| 2.377641 | 0.02245111 | | 3.535534 | 0.01607938 | | 5.303301 | 0.01213414 |
| 2.469221 | 0.02433586 | | 4.045085 | 0.01951271 | | 6.067627 | 0.01561116 |
| | 0.02520761 | | 4.455033 | 0.02320765 | | 6.682549 | 0.02031667 |
| | 0.02337282 | | 4.755283 | 0.02520641 | | 7.132924 | 0.02466012 |
| | 0.01600096 | | 4.938442 | 0.02037584 | | 7.407663 | 0.02259364 |